

江苏大学全英文授课精品课程

# Area of a Surface of Revolution

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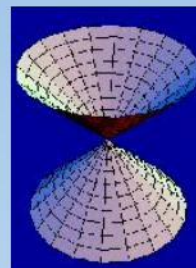
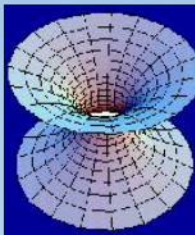
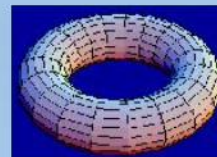
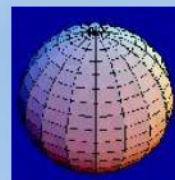
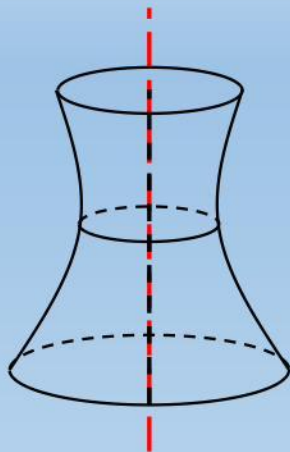
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Jiangsu University

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# 1 What's the Meaning of A surface of Revolution?

Definition : A surface of revolution is formed when a curve is rotated about a line.

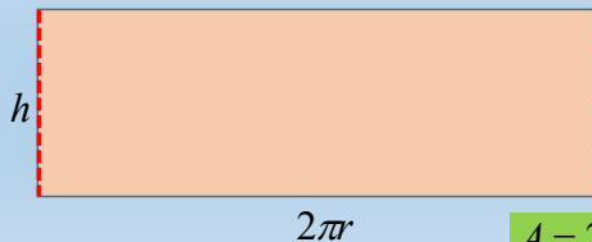
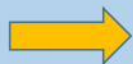
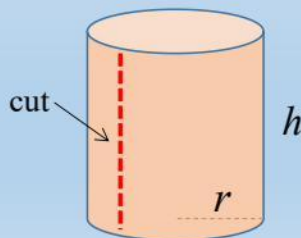


## 2 What's the Meaning of the Area of a Surface of Revolution?

If the surface area is  $A$ , we can imagine that painting the surface would require the same amount of paint as does a flat region with area  $A$

Simple cases:

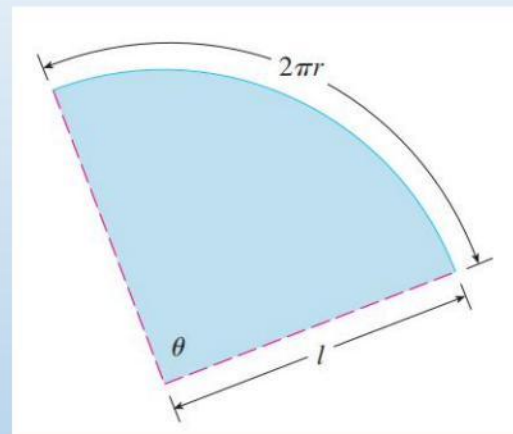
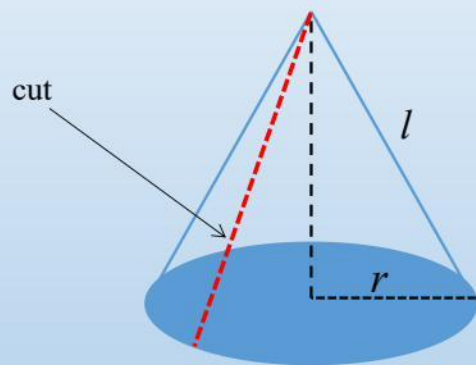
(1) The lateral surface area of a circular cylinder with radius  $r$  and height  $h$



$$A = 2\pi r h$$



(2) The lateral surface area of a circular cone with base radius  $r$  and slant height  $l$



$$\theta = \frac{2\pi r}{l}$$

$$A = \frac{l^2 \theta}{2} = \pi r l$$

The area of the band (or frustum of a cone):

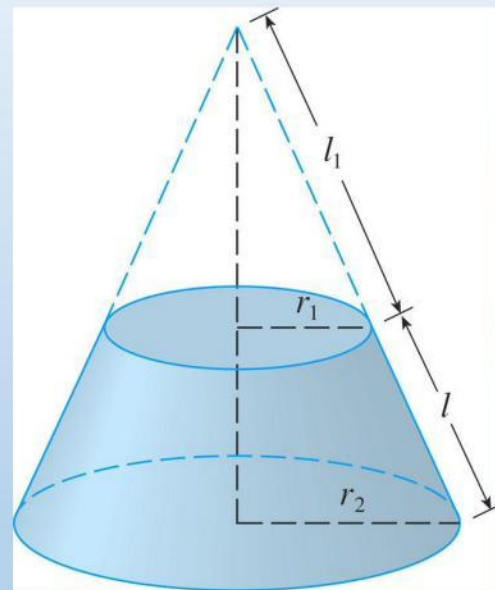
$$A = \pi r_2(l_1 + l) - \pi r_1 l_1 = \pi[(r_2 - r_1)l_1 + r_2 l]$$



$$\frac{l_1}{r_1} = \frac{l_1 + l}{r_2} \text{ (similar triangles)}$$

$$= \pi(r_1 l + r_2 l) = 2\pi r l$$

Here  $r = \frac{r_1 + r_2}{2}$  is the average radius of the band.





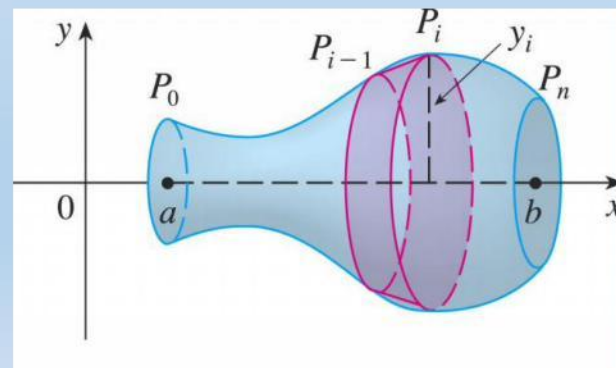
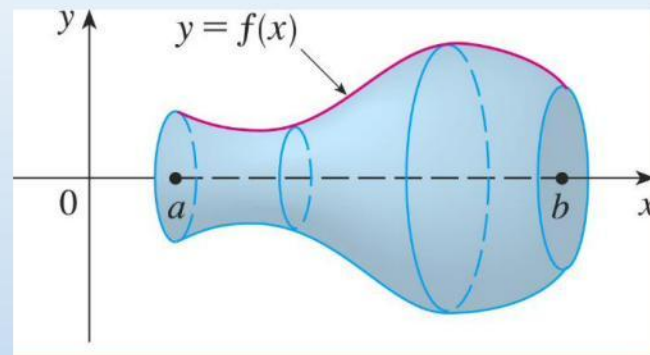
## General cases:

To define its surface,

(1) we divide the interval  $[a, b]$  into  $n$  subintervals with endpoints  $x_0, x_1, \dots, x_n$  and equal width  $\Delta x$ , as we did in determining arc length.

(2) we consider the part of the surface between  $x_{i-1}$  and  $x_i$ , approximated by taking the line segment  $\overline{P_{i-1}P_i}$  and rotating it about the  $x$ -axis.

$$\Delta S_i \approx 2\pi \frac{y_{i-1} + y_i}{2} |P_{i-1}P_i|$$



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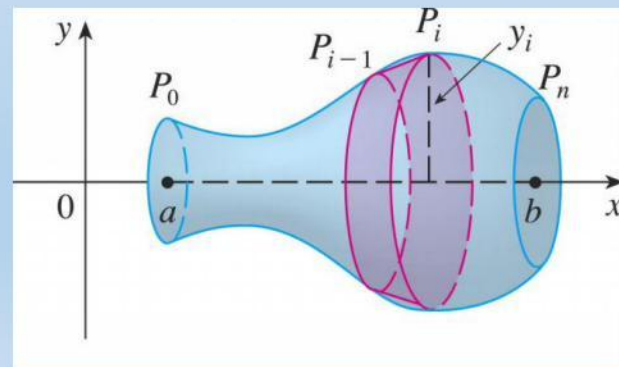
$$y_{i-1} = f(x_{i-1}), y_i = f(x_i), |P_{i-1}P_i| = \sqrt{1 + (f'(x_i^*))^2} \Delta x_i$$

$$\Delta S_i \approx 2\pi \frac{f(x_{i-1}) + f(x_i)}{2} \sqrt{1 + (f'(x_i^*))^2} \Delta x_i$$



$$f(x_{i-1}) \approx f(x_i^*), f(x_i) \approx f(x_i^*), \text{ if } \Delta x_i \text{ is very small}$$

$$\Delta S_i \approx 2\pi f(x_i^*) \sqrt{1 + (f'(x_i^*))^2} \Delta x_i$$



(3) Add all the bands together, we get the approximation to what we think of as the area of the complete surface of revolution is :

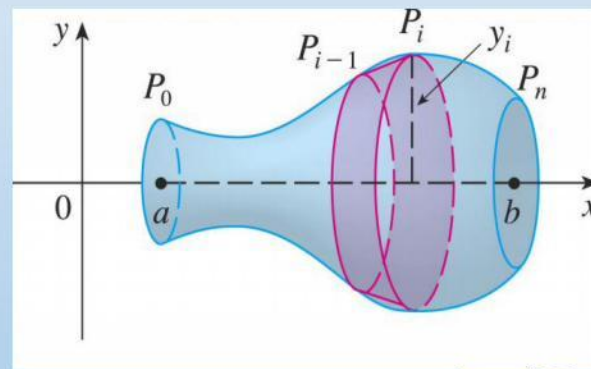
$$S = \sum_{i=1}^n \Delta S_i \approx \sum_{i=1}^n 2\pi f(x_i^*) \sqrt{1 + (f'(x_i^*))^2} \Delta x_i$$

(4) Take limit, we get the actual area :

$$S = \lim_{n \rightarrow \infty} \sum_{i=1}^n 2\pi f(x_i^*) \sqrt{1 + (f'(x_i^*))^2} \Delta x_i$$



$$S = \int_a^b 2\pi f(x) \sqrt{1 + (f'(x))^2} dx$$







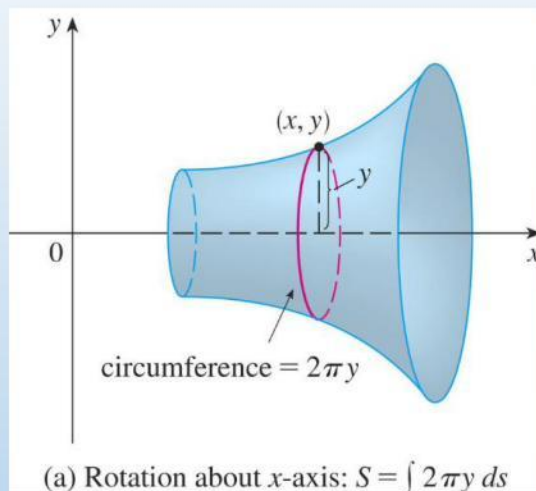
Definition : If  $f(x)$  is positive and has a continuous derivative, we define the surface area of the revolution obtained by rotating the curve  $y = f(x)$ ,  $a \leq x \leq b$ , about the  $x$ -axis as :

$$S = \int_a^b 2\pi f(x) \sqrt{1 + (f'(x))^2} dx = \int 2\pi f(x) ds$$

Definition : If  $g(y)$  is positive and has a continuous derivative, we define the surface area of the revolution obtained by rotating the curve  $x = g(y)$ ,  $c \leq y \leq d$ , about the  $y$ -axis as :

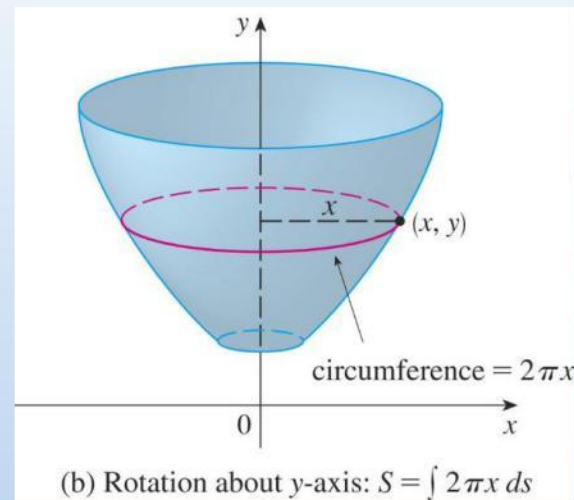
$$S = \int_c^d 2\pi g(y) \sqrt{1 + (g'(y))^2} dy = \int 2\pi g(y) ds$$





$$S = \int_a^b 2\pi f(x) \sqrt{1 + (f'(x))^2} \, dx$$

$$= \int 2\pi f(x) \, ds$$



$$S = \int_c^d 2\pi g(y) \sqrt{1 + (g'(y))^2} \, dy$$

$$= \int 2\pi g(y) \, ds$$





## Remark:

- \* Determine the equation of the curve, which axis did the curve rotated about, If rotated about  $x$  - axis, choose  $x$  as the independent variable and determine the the interval.
- \* Evaluate the derivative of the function.
- \* Use formula of the area of a surface of revolution to calculate.



**Example 1**

Find the area of the surface obtained by rotating the curve

$$y = \sqrt{4 - x^2}, \quad -1 \leq x \leq 1 \text{ about the } x\text{-axis.}$$

**Solution :**

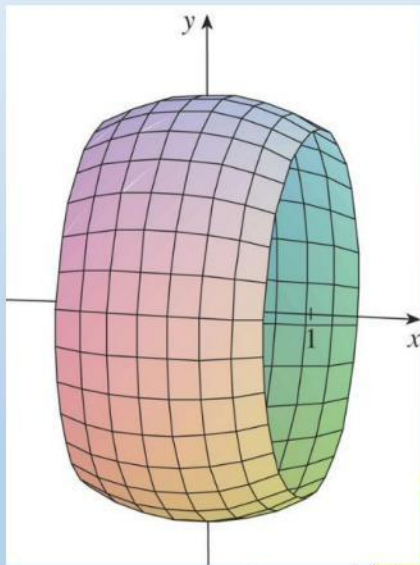
The surface is a portion of a sphere of radius 2.

$$y = \sqrt{4 - x^2} \quad -1 \leq x \leq 1$$

$$y' = \frac{1}{2}(4 - x^2)^{-\frac{1}{2}}(-2x) = \frac{-x}{\sqrt{4 - x^2}}$$

So according to the formula, the surface area is

$$\begin{aligned} S &= \int_{-1}^1 2\pi y \sqrt{1 + (y')^2} dx = 2\pi \int_{-1}^1 \sqrt{4 - x^2} \sqrt{1 + \frac{x^2}{4 - x^2}} dx \\ &= 2\pi \int_{-1}^1 2 dx = 8\pi \end{aligned}$$



## Example 2

Find the area of the surface obtained by rotating the curve  $y = x^2$ ,  $1 \leq x \leq 2$  about the  $y$ -axis.

### Solution :

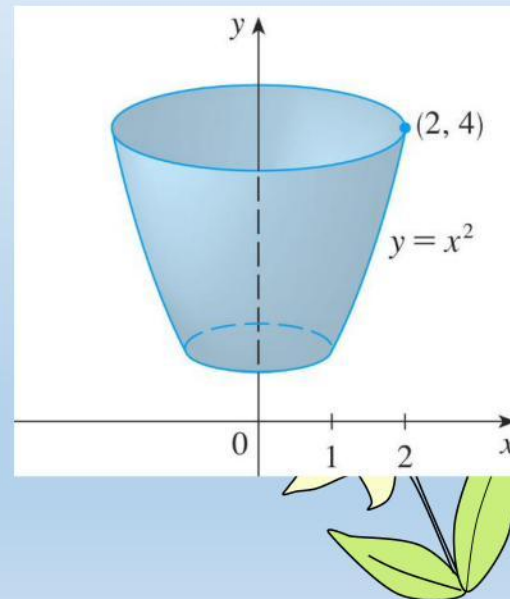
Notice : rotating the parabola about  $y$  - axis

independent variable is  $y$ , and  $y \in [1, 4]$

$$x = \sqrt{y} \quad \text{and} \quad x' = \frac{1}{2}y^{-\frac{1}{2}}$$

So according to the formula, the surface area is

$$\begin{aligned} S &= \int_1^4 2\pi x \sqrt{1 + (x')^2} dy = 2\pi \int_1^4 \sqrt{y} \sqrt{1 + \frac{1}{4y}} dy \\ &= \pi \int_1^4 \sqrt{4y + 1} dy \\ &= \frac{\pi}{12} (17\sqrt{17} - 5\sqrt{5}) \end{aligned}$$







### Example 3

Find the area of the surface generated by rotating the curve  $y = e^x$ ,  $0 < x < 1$  about  $x$ -axis.

#### Solution :

Since about  $x$  - axis, so we choose  $x$  as the independent variable.

and the range of  $x$  is  $(0,1)$ . Since  $y = e^x$ , we have  $\frac{dy}{dx} = e^x$

So according to the formula, the surface area is

$$S = \int_0^1 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = 2\pi \int_0^1 e^x \sqrt{1 + e^{2x}} dx$$

$$= 2\pi \int_1^e \sqrt{1 + u^2} du \quad \text{--- } u = e^x$$

$$= 2\pi \int_{\pi/4}^{\arctan e} \sec^3 \theta d\theta \quad \text{--- } u = \tan \theta$$

$$= \pi [e\sqrt{1+e^2} + \ln(e + \sqrt{1+e^2}) - \sqrt{2} - \ln(\sqrt{2} + 1)]$$





## Exercises

Find the area of the resulting surface.

(1)  $y = \sqrt[3]{x}$ ,  $1 \leq y \leq 2$  rotated about  $y$ -axis

(2)  $y = \sqrt{1+4x}$ ,  $1 \leq x \leq 5$  rotated about  $x$ -axis

(3)  $y = \frac{x^3}{6} + \frac{1}{2x}$ ,  $\frac{1}{2} \leq x \leq 1$  rotated about  $x$ -axis

(4)  $y = \ln(x+1)$ ,  $0 \leq x \leq 1$  rotated about  $y$ -axis

